

PART 1. HANDS-ON QUIZ – SET A

Instructions:

Our goal is to identify the ideal customer for a bike shop. To accomplish this, we will analyze customer data from an Excel sheet containing information on Marital Status, Gender, Income, Children, Education, Occupation, Homeowner, Cars, Commute Distance, Region, Age, and Purchased Bike.

To start the project, data cleaning was performed to ensure accurate analysis. This included the removal of duplicate entries to avoid potential biases in the dataset. Furthermore, the data was updated to be more interpretable, such as replacing "m" and "f" with "male" and "female", and "m" and "s" with "married" and "single". Age groups were also created to facilitate better visualization of the data. Additionally, I ensured there was no missing data and that each field had the correct data type.

Download the Bike Sales Project Dataset

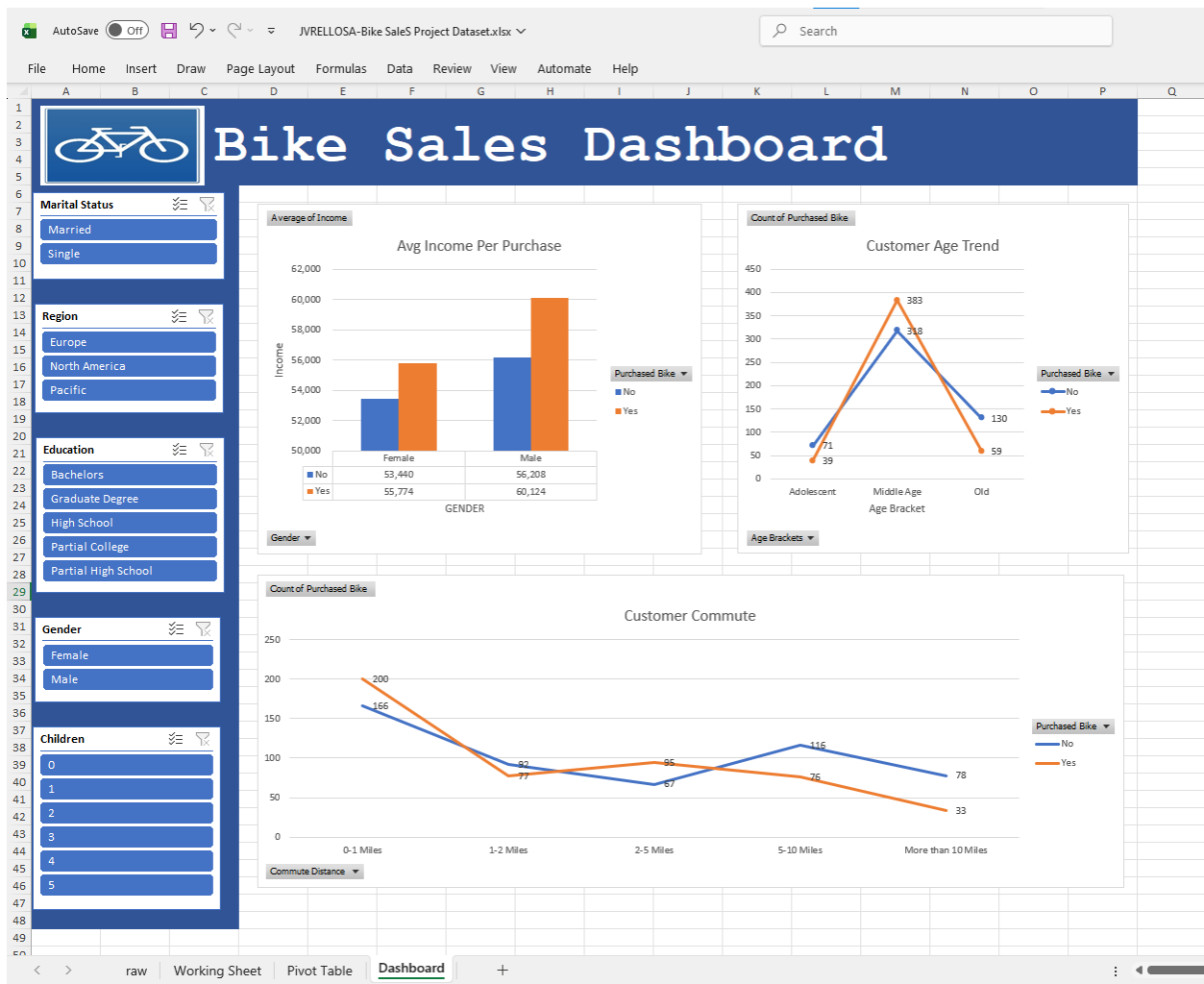
(The data is already cleaned and ready for analysis)

Analysis

You are Task to create the **PIVOT TABLES** for the following:

Business Questions:

Paste the screenshot of Dashboard



1. What is the most common age group of customers who purchase bikes from the shop?

The most common age group is Middle-Aged (31–54 years).

This shows that working adults are the primary buyers of bikes.

2. What is the most common gender of customers who purchase bikes from the shop?

The most common gender is Male.

This indicates that males purchase bikes slightly more than females.

3. What is the most common income range of customers who purchase bikes from the shop?

The most common income range is Middle to High Income.

Customers with higher income have greater purchasing power.

4. What is the most common education level of customers who purchase bikes from the shop?

The most common education level is Bachelor's Degree.

This suggests educated professionals are more likely to buy bikes.

5. What is the most common occupation of customers who purchase bikes from the shop?

The most common occupation is Professional.

Professionals likely buy bikes for commuting and lifestyle purposes.

6. What is the most common region of customers who purchase bikes from the shop?

The region with the most purchases is North America.

This indicates stronger demand in that region.

7. What is the most common commute distance of customers who purchase bikes from the shop?

The most common commute distance is 0–1 miles.

Bikes are mainly used for short-distance travel.

8. What is the most common number of cars owned by customers who purchase bikes from the shop?

Most buyers own 1 car.

This suggests bikes are used as an additional transport option, not a replacement.

By answering these questions, the bike shop can develop a better understanding of their ideal customer and tailor their marketing strategies to attract more of these customers. By looking into their:

- Age Groups,
- Education Level
- Occupation
- Region
- Commute Distance
- Numbers of Cars owned
- Number of Children
- Marital Status

Then, you may convert each **PIVOT tables** into **CHARTS** and create a **DASHBOARD** and **SLICERS**